BIOMATERIALS

Soft strain-insensitive bioelectronics featuring brittle materials

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Advancing electronics to interact with tissue necessitates meeting material constraints in electrochemical, electrical, and mechanical domains simultaneously. Clinical bioelectrodes with established electrochemical functionalities are rigid and mechanically mismatched with tissue. Whereas conductive materials with tissue-like softness and stretchability are demonstrated, when applied to electrochemically probe tissue, their performance is distorted by strain and corrosion. We devise a layered architectural composite design that couples strain-induced cracked films with a strain-isolated out-of-plane conductive pathway and in-plane nanowire networks to eliminate strain effects on device electrochemical performance. Accordingly, we developed a library of stretchable, highly conductive, and strain-insensitive bioelectrodes featuring clinically established brittle interfacial materials (iridium-oxide, gold, platinum, and carbon). We paired these bioelectrodes with different electrochemical probing methods (amperometry, voltammetry, and potentiometry) and demonstrated strain-insensitive sensing of multiple biomarkers and in vivo neuromodulation.

Interfacing electronics with biological tissues is the foundation of probing and actuating biological systems. Reliable interaction between biological tissues and electronics requires the use of materials that support underlying electrochemical and electrical processes and satisfy tissue-imposed mechanical constraints (*1*, *2*). The electrochemical process involves electron transfer and accumulation at the bioelectronics-tissue interface for in situ stimulation (for example, neuromodulation) and sensing (for example, biomarker molecules), necessitating the use of suitable interfacial materials. The electrical process involves electron transport for signal routing and setting intended operating points (for example, voltage), which requires the use of highly conductive materials (*3*). From the mechanics standpoint, materials that mimic tissue properties such as stretchability (~20 to 75%) and compliance (Young's modulus, E, on the order of kilopascals to megapascals) are needed to establish conformable contact with tissue for high fidelity sensing and stimulation. This requirement is critical for minimizing disruption of tissue function as well as the risk of implant complications that result from scar formation, lead fracture, and tissue injury (*1*). However, the electrochemical and electrical performance of these materials may be affected by the strain originating from the movement and complex topography of the tissue.

Clinically established interfacial materials with high electrochemical performance for bioelectronics-tissue interfaces (such as noble metal–based ones) are all rigid (E > 1 GPa) and brittle (fracture strain < 1%). Recent advances in soft and stretchable conductors have enabled the development of strain-resilient devices with tissue-like mechanical properties, which maintain their electrical connection despite being extensively stretched (~800%) (*4*, *5*). Although these strain-resilient devices are suitable for the construction and integration of soft, stretchable, solid-state electronics, when they are applied to electrochemically probe the tissue environment (involving ionic biofluid surrounding), their performance is distorted by strain- or corrosion-related issues. The former issue manifests in devices with relatively low intrinsic conductivity (such as graphene and PEDOT:PSS: 1 to 10^3 S/m), in which strain-induced changes to their resistance cause substantial deviation from the intended operating points (such as, activation voltage), subsequently corrupting the reactions at the interface (*6*–*8*). The latter issue is especially observed in devices with highly conductive materials that rapidly oxidize upon exposure to the surrounding solution (such as silver-based nanomaterials), despite the application of surface modification [prone to surface defects or pores (*9*)]. As such, this group of devices cannot sustain the wide voltage range required to perform electrochemical reactions (*5*, *10*).

We devised a layered architectural composite design that centers on decoupling the bioelectronic materials configuration into an interfacial element for electron transfer and an interconnection element for electron transport. This design allows for the exploiting and coupling of surface channel cracks (within the brittle interfacial element) and anisotropic out-of-plane/in-plane electron conduction (within the interconnection element) to eliminate strain effects on device performance. The design further allows for a broad selection for the interfacial materials, including brittle noble metal–based materials. Following this approach, we created a functionally diverse library of thin (~140 μm), soft (~10 MPa) (fig. S1), stretchable (>150%), highly conductive (~1 ohm/sq) and strain-insensitive bioelectrodes (SIBs).

Our fabrication strategy (Fig. 1A) is based on (i) constructing the tissue interfacing element as a thin film with favorable electron transfer characteristics, (ii) constructing the interconnection element by using intrinsically stretchable and highly conductive silver nanowire (AgNW)–based traces, which are inlaid in the surface layer of a soft rubbery matrix [poly(urethane acrylate), PUA], and (iii) seamlessly integrating the two elements by means of a thin adhesive and anisotropically conductive film (ACF), which features isolated conductive particles embedded in a stretchable matrix to facilitate anisotropic out-of-plane electrical conduction. A variety of thin-film materials, including those that are brittle but have high electrochemical performance [gold (Au) platinum (Pt), and iridium oxide (IrOx)], can be built atop the ACF-AgNW-PUA layers through standard thin-film deposition techniques (Fig. 1B).

Fundamentally, our SIB's strain insensitivity is controlled by three layer-specific strain energy dissipation mechanisms (Fig. 1, C and D, and fig. S2): (i) crack channeling of brittle interfacial thin film, (ii) strain isolation of the out-of-plane conductive pathway in the ACF layer, and (iii) realignment of in-plane AgNW networks. Interfacial channel cracks are exploited as means of tensile strain energy release. Each cracked fragment experiences minimal strain, while remaining electrically connected to the system, preserving the overall active surface area of the SIB. As the electrical bridge between the cracked interfacial thin film and the AgNW-PUA layer, the ACF renders strain-insensitive out-of-plane electrical conduction,

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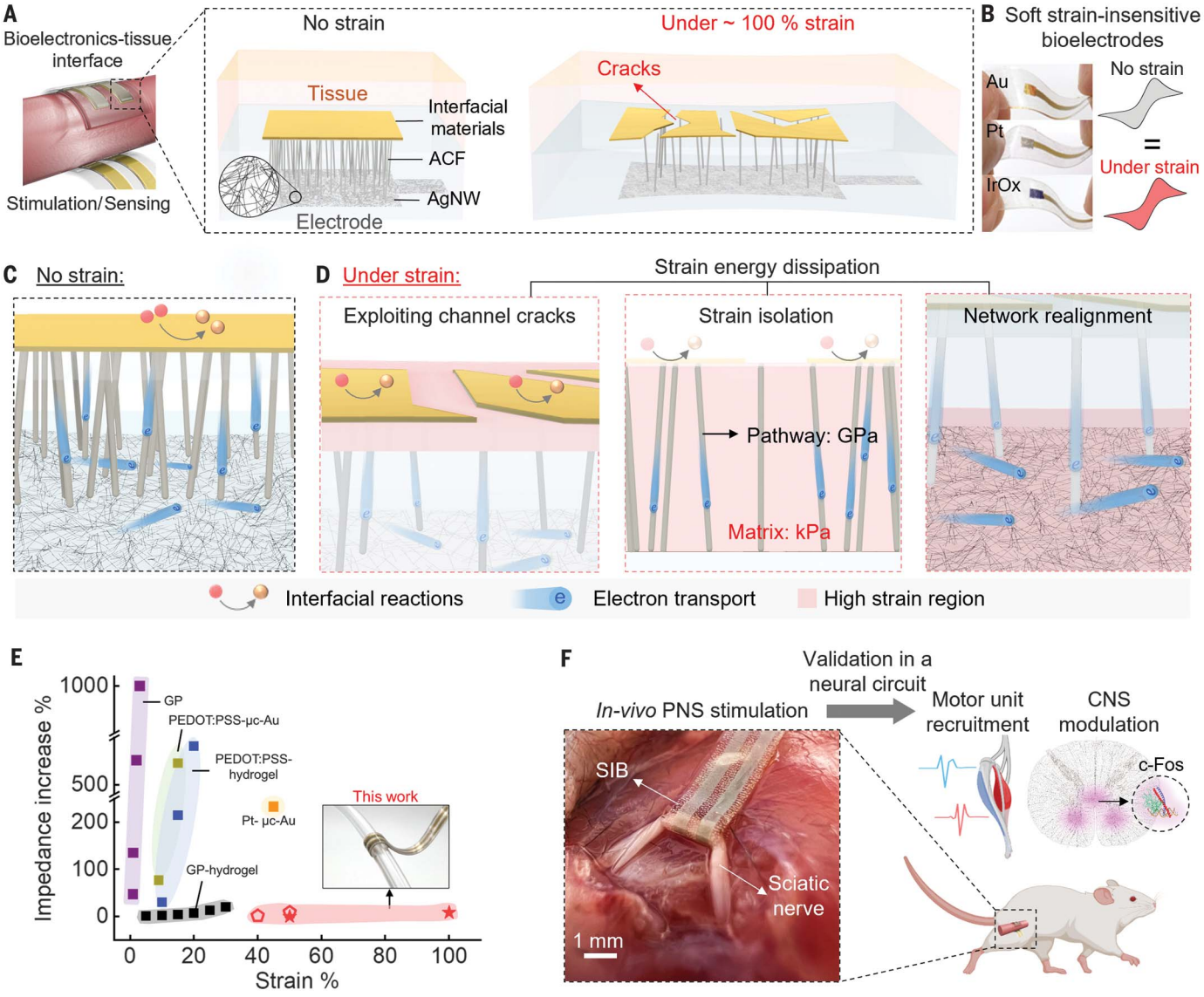


Fig. 1. Soft strain-insensitive bioelectrode (SIB): architecture, strain dissipation mechanisms, and applications. (**A**) Illustration of the SIB as a bioelectronics-tissue interface under strain. (**B**) Brittle noble metal–enabled SIBs for high fidelity stimulation and sensing under strain. (**C**) Schematic presentation of the SIB's underlying electron transfer and out-of-plane and in-plane electron transport processes. (**D**) Strain energy dissipation mechanisms within each layer of the SIB. (**E**) Ashby diagram of electrochemical impedance (at 1 kHz) increase versus tensile strain (*7*, *8*, *10*, *12*, *13*). Au-SIB and IrOx-SIB are indicated with star and pentagon markers, respectively. (**F**) A representative SIB interfacing the sciatic nerve of a mouse for nerve stimulation to achieve synchronized motor unit recruitment and CNS modulation, manifested as c-Fos expression.

because the rigid conductive microparticles (approximately gigapascals) are strain-isolated by the surrounding soft matrix (approximately kilopascals). The AgNWs in the bottom layer form in-plane conductive percolation networks that are strongly anchored onto the soft PUA substrate by means of in situ cross-linking. This strong binding enables network realignment to release strain energy (*11*).

The strain sensitivity of the electrical and electrochemical performances of the SIBs is compared with those obtained by the state-of-the-art stretchable bioelectrodes (Fig. 1E, fig. S3, and table S1) (*7*, *8*, *10*, *12*, *13*). We applied our SIBs to perform high-fidelity sensing of multiple biomarkers and neurostimulation under strain. We validated the in vivo operation of our SIBs in the context of a neural circuit. Specifically, we interfaced the bioelectrodes, featuring a brittle IrOx layer, with the sciatic nerve and verified the modulation of the bidirectional and interconnecting central nervous system (CNS) and motor unit (within the neural circuit) through peripheral nerve stimulation (PNS).

To visualize and quantitatively characterize the SIBs' strain insensitivity, we correspondingly used an electrochemical deposition–based staining method, electrical impedance spectroscopy (EIS), and cyclic voltammetry (CV). We characterized Au-deposited SIBs (Au-SIBs) as model test devices over a range of tensile strain conditions. The devised electrochemical staining method enables the spatial mapping of the electrical connection of the interfacial thin film. It uses a standard electrodeposition setup to deposit a conductive material (in this application, Pt) onto a test electrode under strain, in which only the electrode surface regions that remain electrically connected to the power source support electrodeposition (Fig. 2A). As a result, the electrochemically deposited regions

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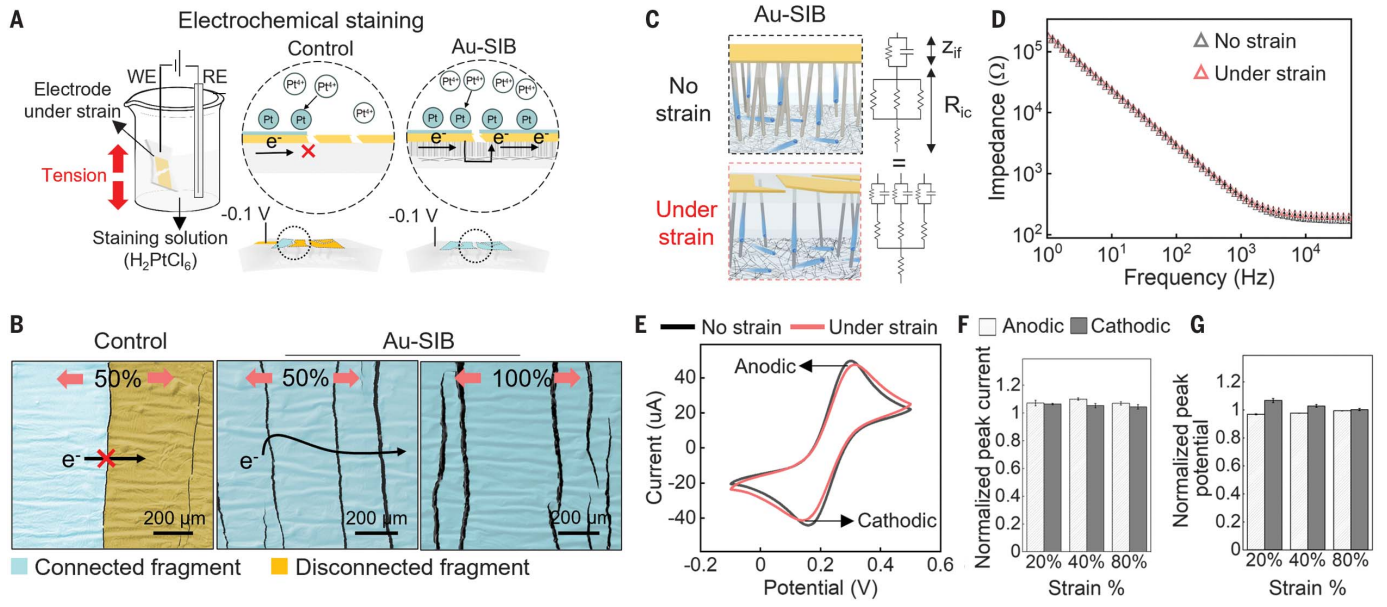


Fig. 2. Visualization and electrochemical characterization of Au-SIB performance under strain. (**A**) Schematic of the introduced electrochemical staining method. (**B**) Colored SEM images of the electrochemically stained control (Au, ACF, and PUA) electrode and Au-SIB under strain. They illustrate the electrical disconnection of a cracked fragment in the control electrode, whereas the Au-SIB's cracked fragments remained electrically connected. (**C**) Corresponding Randles model schematics of unstrained and strained or cracked SIBs. Interconnect resistance: R_{ic}. Interfacial impedance: Z_{if}. The SIB design enables the formation of parallel pathways for electron transfer and transport and maintenance of the active surface area under strain. (**D**) EIS comparison of Au-SIBs: unstrained versus 100% tensile-strained (n = 3 measurements by the same device for each strain condition). P = 0.12 comparing strained and unstrained measurement groups. Error bars indicate SEM. (**E**) CV comparison of Au-SIBs: unstrained versus 100% tensile-strained (in $K_3Fe(CN)_6$ solution). (**F** and **G**) The anodic and cathodic peak positions (F) and heights (G) of Au-SIBs under 20% through 80% strain (normalized to unstrained values; N = 3; p values 0.053 to 0.745; error bars indicate SEM).

("stained" regions) of the test electrodes exhibit a substantial imaging contrast under scanning electron microscopy (SEM) as compared with the nondeposited regions (fig. S4). As illustrated in Fig. 2B, the cracked fragments of a representative Au-SIB were fully "stained," indicating the preservation of their conductivity under varying large strain levels, in contrast to that of a control Au-elastomer electrode.

We characterized the cracked fragments by means of SEM imaging and analyzed the results with the aid of a dedicated statistical model (supplementary text S1). The results indicate that >99% of the brittle interfacial layer's surface area remains connected (fig. S5). Our cyclic stretching studies also demonstrate that the Au-SIB's crack density does not increase, despite repeated stretching (>10,000 cycles at 20% strain) (fig. S6A), and that the device's low resistance is minimally impacted during dynamic stretching (~2 ohm variation) (fig. S6B and C).

EIS and CV characterization results indicate that the changes to the SIB system's underlying electrochemical and electrical components (modeled in Fig. 2C) are negligible under strain. The impedance spectrum of representative 100% strained SIBs exhibited statistically identical curves in impedance magnitude (1 Hz to 100 kHz), as compared with their unstrained states (Fig. 2D and fig. S7). The SIB's electrochemical impedance has minimal change after 1000 cycles of tension (fig. S8). Furthermore, the CV plots of the unstrained and strained SIBs exhibit nearly identical features across 20 to 100% strain (Fig. 2, E to G). Without strain accommodation, tissue-induced strain causes a substantial increase in the interconnect resistance to the point of open circuit, which corrupts the electrode's electrical and electrochemical characteristics that manifest as as shifting and distortion of the EIS and CV curves, respectively (fig. S9).

Compared with bare AgNW-PUA, Au-SIB is more electrochemically stable (fig. S10). The underlying AgNW-PUA layer of the SIB does not experience galvanic corrosion in the presence of cracking. This contrasts with the case of Au-deposited AgNW-PUA electrodes, which galvanically corrode because of the crack-induced exposure of the AgNWs (fig. S11). The SIB's corrosion resistance indicates the protective role of the hydrophobic ACF in the prevention of AgNW oxidation and the release of cytotoxic Ag ions (useful for prolonged and biocompatible in vivo operations).

A diverse library of interfacial materials beyond Au can be simply incorporated into the SIB structure with standard thin film deposition techniques and paired with different application-specific electrochemical probing methods that involve wide ranging voltage and frequency conditions (Fig. 3A). We deposited Pt, carbon, and IrOx onto ACF as examples of commonly used brittle interfacial materials, leading to a series of SIBs with strain-insensitive interfacial electrochemical reaction features (validated with CV characterization) Fig. 3B). We applied these electrodes (exhibiting minimal device-to-device variation) (fig. S12) under large strain to carry out amperometry, voltammetry, and potentiometry, representing distinct electron transfer processes [Fig. 3, C (i, ii), D (i, ii), and E (i, ii)].

The choice of Pt as an interfacial material was motivated by its common use as a hydrogen peroxide (H_2O_2)–sensing layer in oxidase-based enzymatic biosensors. As shown in Fig. 3C, and figs. S13A and S14A, the corresponding amperometric responses of the Pt-SIB at its relaxed and tensile states were similar for different concentrations of H_2O_2. These results indicate that the faradaic current was minimally affected by the strain despite the occurrence of the cracks. We also observed strain-insensitive H_2O_2-sensing when using Au as an interfacial material (fig. S15).

The choice of carbon as an interfacial material was motivated by its common use in the quantification of low-concentration electroactive biomarkers. We investigated the effect of strain on the response of the carbon-SIB electrodes in the context of differential pulse

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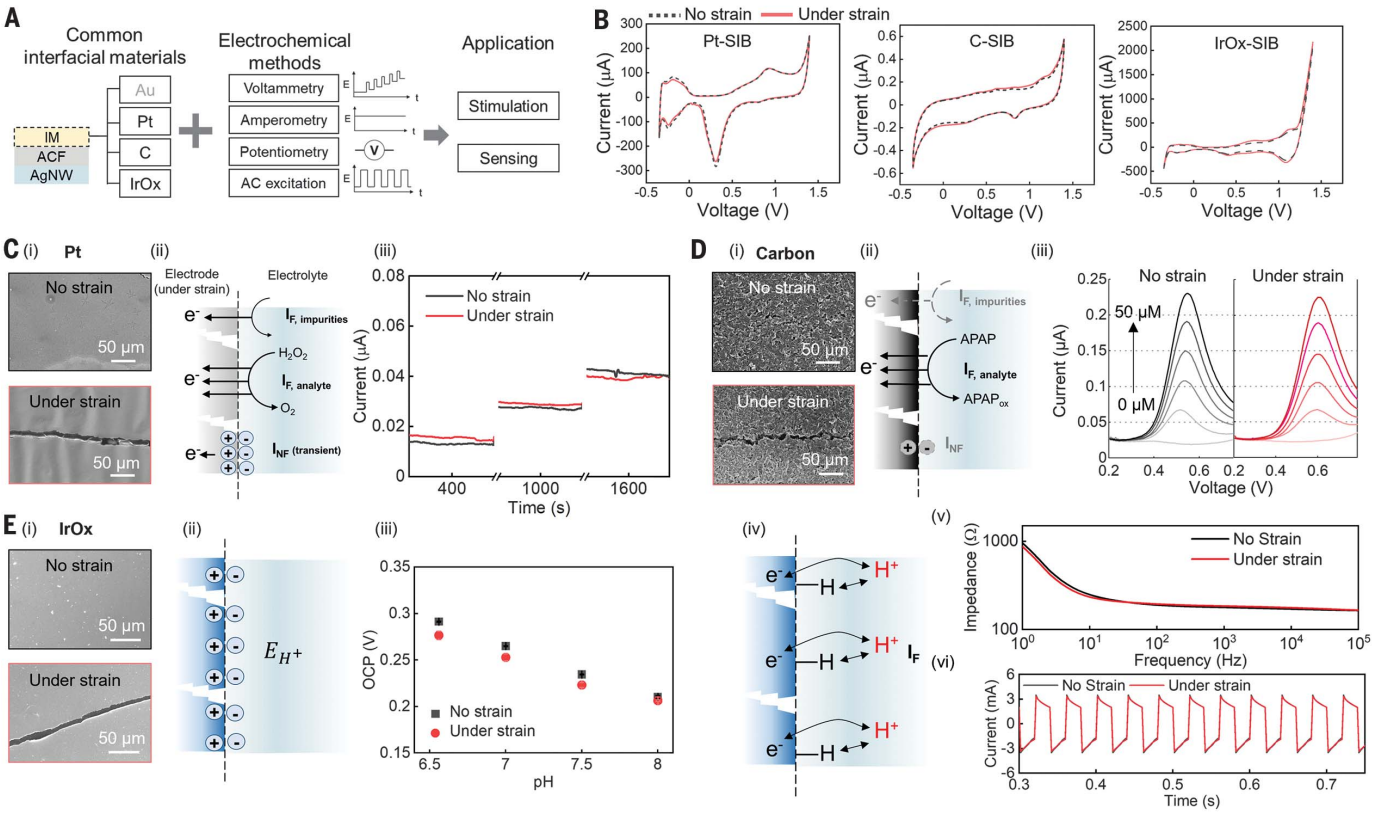


Fig. 3. A library of SIBs featuring brittle interfacial materials for electrochemical sensing and stimulation. (**A**) Pairing of SIBs with different electrochemical probing methods for diverse applications. IM represents the interfacial material layer. (**B**) CV characteristics of Pt-SIB, Carbon-SIB, and IrOx-SIB at 40% tensile strain versus unstrained state. (**C**) Pt-SIB characterization: (i) SEM images of the unstrained and 40% strained Pt-SIBs, (ii) schematic of the amperometric H_2O_2 sensing process, and (iii) Pt-SIB's H_2O_2 responses (0, 50, and 100 μM) at 40% tensile strain versus unstrained. (**D**) Carbon-SIB characterization: (i) SEM images of the unstrained and 100% strained carbon-SIBs, (ii) schematic of the DPV acetaminophen (APAP) sensing process, and (iii) carbon-SIB's APAP responses at 100% tensile strain versus unstrained. (**E**) IrOx-SIB characterization: (i) SEM images of the unstrained and 40% strained IrOx-SIBs, (ii) schematic of the pH sensing process, (iii) IrOx-SIB's pH open circuit potential (OCP) responses at 40% tensile strain versus unstrained, (iv) schematic of the faradaic process for neurostimulation, and (v, vi) corresponding EIS and ex vivo neurostimulation characterization of the IrOx-SIBs at 40% tensile strain versus unstrained.

voltammetry (DPV) for the detection of acetaminophen. As shown in Fig. 3D (iii) and fig. S13B, the corresponding DPV peak current measured by the electrode at its relaxed versus tensile states were similar for different concentrations of acetaminophen. The strain-induced shift in reduction-oxidation (redox) potential was minimal (~0.1 V).

The choice of IrOx as an interfacial material was motivated by its substantial yet unrealized full potential in bioelectronics. IrOx is not only biocompatible but also facilitates fast and reversible redox reactions with surrounding hydrogen ions (*14*), enabling large and reversible current injection. The latter point in particular supports IrOx's superiority over noble metals such as Au and Pt for tissue bioelectronics applications such as pH sensing and neural stimulation. However, because of its intrinsic brittleness, IrOx cannot be easily engineered as an inherently stretchable electrode. Our fabrication strategy enabled simple engineering of IrOx-SIBs where the tight bonding of the IrOx film (supporting electron transfer) to the underlying layer (supporting electron transport) ensured robust device operation despite IrOx's brittleness. The fabricated IrOx-SIB possesses a charge storage capacity of ~12 mC/cm^2 (fig. S16), which is similar to the charge storage capacity of standard clinical stimulation electrodes (*15*).

We investigated the IrOx-SIB for potentiometric pH sensing [Fig. 3E (ii)] and AC excitation in the form of biphasic pulses, for neurostimulation [Fig. 3E (iv)]. The open-circuit potential measurements indicate that the IrOx-SIB responses are correlated with the tested solution's pH levels, and that the electrode responses remain unperturbed under both static and dynamic tensile states [Fig. 3E (iii) and fig. S13C and fig. S14B]. As plotted in Fig. 3E (v) and fig. S17, the EIS measurements performed with our unstrained versus 40% strained IrOx-SIBs were similar, illustrating its strain insensitivity over a wide frequency range and informing its utility for both low- and high-frequency stimulation settings such as neuromodulation. Our studies in ex vivo neurostimulation characterization demonstrate the SIB's robustness in delivering stable current under both static and dynamic tensile conditions [Fig. 3E (vi) and fig. S14C].

As summarized in Fig. 4A, the IrOX-SIB presents electrochemical and electrical properties nearly identical to those of clinically used rigid cuff electrodes, while exhibiting tissue-like mechanical properties. The neural tissue–like mechanical characteristic of IrOx-SIB is advantageous over existing implantable electrodes that use rigid materials for neuromodulation because it potentially minimizes scar formation and immune response while rendering efficient stimulation even under deformation (*1*). To evaluate the performance of IrOx-SIBs during in vivo stimulation, we interfaced these bioelectrodes with the sciatic nerve to modulate the bidirectional and interconnecting spinal cord and muscles within the neural circuit (Fig. 4B). Accordingly, after validating its biocompatibility through cellular viability studies (fig. S18), the IrOx-SIB was applied in living mouse studies to deliver

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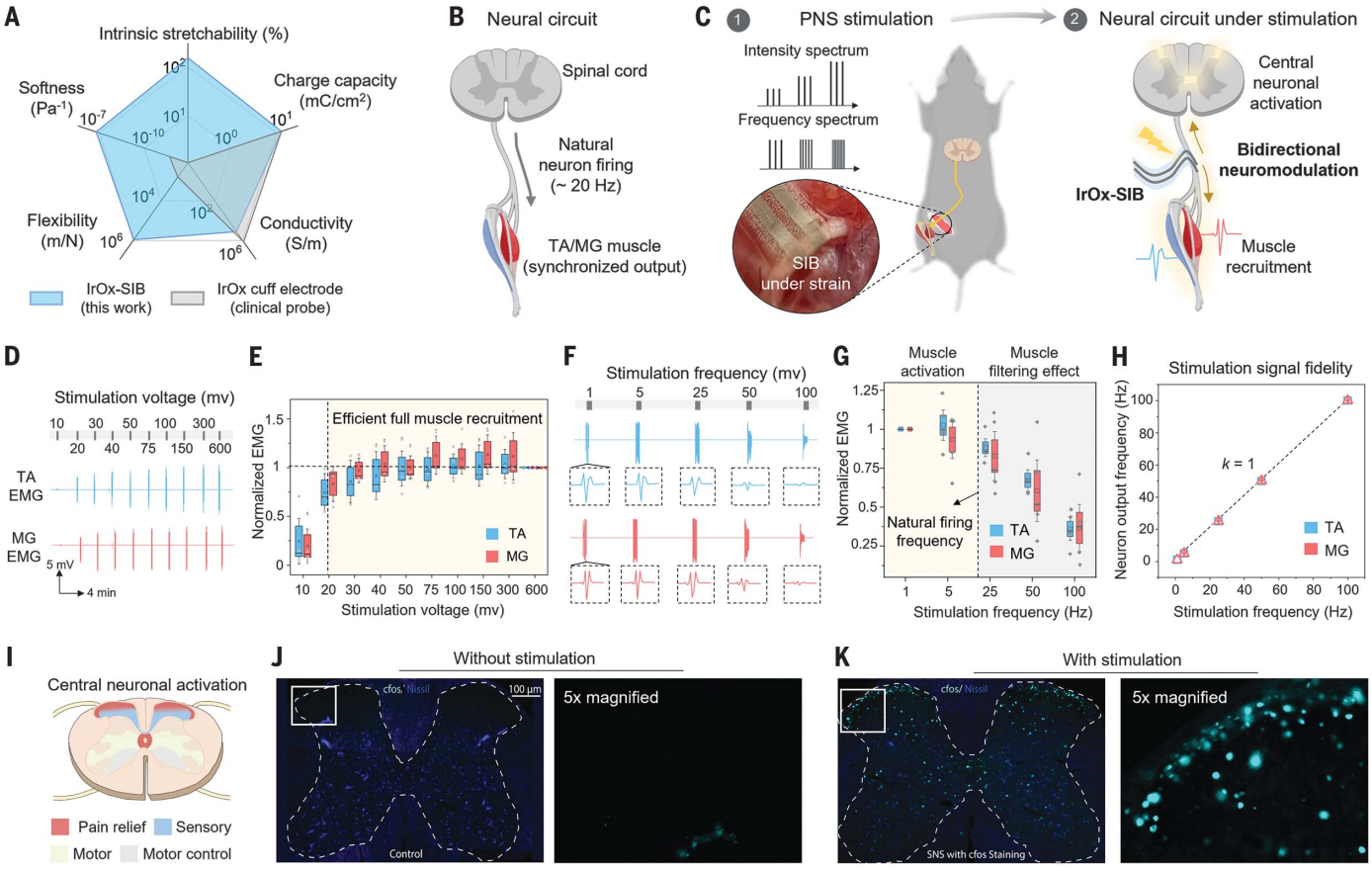


Fig. 4. In vivo neuromodulation through stimulation of the sciatic nerve by IrOx-SIB under deformation. (**A**) Radar chart of electrochemical, electrical, and mechanical properties of IrOx-SIB versus a standard IrOx cuff electrode. (**B**) Illustration of the neural circuit of a mouse. (**C**) In vivo evaluation of IrOx-SIB neuromodulation performance in an established neural circuit model. PNS responses were monitored through TA and MG recordings and CNS neuromodulation was evaluated through c-Fos staining of the lumbar spinal cord. (**D** to **G**) Sciatic nerve stimulation with different stimulation voltages at 1 Hz [(D) and (E)] and frequencies at 100 mV [(F) and (G)], and the correspondingly induced TA and MG muscle EMG recordings [(D) and (F)] and their respective normalized amplitudes [(E) and (G)]. (**H**) Comparison of the frequency of TA and MG neural signal outputs captured in G versus the stimulation frequency. (**I**) Schematics of regions determined by their neural functions within the spinal cord. (**J** and **K**) c-Fos expression patterns within lumbar spinal cord with (K) and without (J) sciatic nerve stimulation. Whole spinal cord staining (c-Fos single channel) images and zoomed-in views of the selected regions are shown. Dark blue color represents nissil staining and cyan color represents c-Fos staining.

clinically relevant voltage and frequency stimulation conditions.

We concurrently recorded the real-time electromyography (EMG) of the bilateral anterior tibialis (TA) and medial gastral (MG) muscles innervated by the sciatic nerve (Fig. 4C). To validate CNS neuromodulation, the subsequently expressed c-Fos protein levels (induced through sciatic nerve stimulation) in the lumbar spinal cord were labeled with immunofluorescent staining (Fig. 4C). Sciatic nerve stimulation with increasing voltage levels (at 1 Hz of frequency) led to the increase and following saturation of the TA and MG EMG intensities (Fig. 4, D and E). This trend indicates an increase in the synchronized recruitment of the available motor units, up to the point of full recruitment, which is achieved at potential levels as low as ~20 mV, thus illustrating efficient recruitment (*8*). As shown in Fig. 4, F and G, our studies involving sciatic nerve stimulation with excitation frequencies spanning from 1 to 100 Hz (at 100 mV) revealed the low-pass filtering characteristic (with cut-off frequency at ~20 Hz) of the muscle contraction (*16*). Despite being strained, our device induced TA and MG EMG with similar levels of signal-to-noise ratio (SNR; ~30 dB) as those produced by conventional electrodes (~10 to 38 dB at ~1 Hz) in unstrained settings (*17*, *18*).

As shown in fig. 4H, there is a one-to-one ratio between the applied stimulation frequencies and the recorded TA and MG muscle contraction frequencies. These results demonstrate the fidelity of the neural stimulation and the robustness of the strained SIB in supporting undistorted electron transfer and transport processes. From the standpoint of CNS neuromodulation (Fig. 4I), the c-Fos proteins are observably expressed in response to the sciatic nerve stimulation (Fig. 4K) (nonstimulated control shown in Fig. 4J). Furthermore, our staining results indicate the spatial concentration of the c-Fos expression in the dorsal horn of the spinal cord, which is related to the neural processing of sensory information such as pain.

We harnessed the superior electrochemical stability of diverse interfacial materials and the electrical conductivity of silver nanowire without being restricted by their inherent limitations in stretchable bioelectronic settings (interfacial materials' crack onset strain and silver nanowire's electrochemical instability). These materials were combined through the use of a strain-insensitive and anisotropic ultraconductive film. We simultaneously achieved strain insensitivity, high conductivity, and electrochemical stability with a diverse array of interfacial materials. On a broader level, by

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unlocking mechanical constraints, the design principle allows for the integration of materials in ways that may harness their superior properties in different domains. This allows for the attainment of maximum achievable performance offered across constituent materials.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Supplementary Text
Figs. S1 to S18
Table S1
References (*19*–*30*)

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Staying soft and conductive under strain

Most electrically conductive materials tend to be stiff and brittle, whereas human tissue is soft and compliant. It is thus a challenge to make conductive biomaterials that are sufficiently compliant but do not show a loss or distortion in performance. Zhao *et al*. used a three-layer design to couple strain-induced cracked films with a strain-isolated conductive pathway (see the Perspective by Rafeedi and Lipomi). Upon an initial prestrain to 100%, the brittle solid film on top cracks to dissipate the strain energy. However, this cracking permits a type of parallel, interconnected charge transport in which the charge carriers move between the layers to circumvent the cracks. —MSL

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